

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

NAME: U. Somesh Kumar

REGISTER NUMBER: 192425019

COURSE OUTCOME COVERED

CO4: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 30%

Dr G A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Mock Interview

- 1. A machine learning engineer applies Reinforcement Learning to solve sequential decision-making problems. Explain how an agent learns optimal behaviour through interaction with the environment.**

ANS: - Reinforcement Learning (RL) is a learning approach in which an agent learns optimal behavior by interacting with an environment.

Working Process:

1. **State:** The agent observes the current state of the environment.
2. **Action:** Based on its policy, the agent selects an action.
3. **Reward:** The environment provides a reward or penalty for the selected action.
4. **Next State:** The environment changes to a new state.

5. Learning: The agent uses the received reward to update its policy or value function.
6. Optimization: Through repeated interactions, the agent learns actions that maximize the cumulative reward.

Example:

A robot learning to reach a destination receives positive rewards for moving closer and penalties for hitting obstacles. Over time, it learns the safest and shortest path.

Flow:

State → Action → Environment → Reward + Next State → Policy Update → Improved Action

Conclusion:

RL learns optimal behavior through trial and error by maximizing long-term cumulative rewards.

2. **If you are asked to design a Reinforcement Learning solution for a real-time system such as ride-sharing or delivery optimization, explain how you would formulate the problem using a Markov Decision Process.**

ANS: - A real-time ride-sharing or delivery system can be formulated as a Markov Decision Process (MDP) using five main components:

1. State (S)

Represents the current situation of the system.

Examples:

- Vehicle location
- Number of available drivers
- Current traffic conditions
- Pending ride/delivery requests
- Battery or fuel level

2. Action (A)

Represents the decision taken by the RL agent.

Examples:

- Assign a driver
- Select a delivery route
- Accept/reject a request
- Reposition a vehicle

3. Reward (R)

Measures the quality of the decision.

Examples:

- Higher reward for faster delivery
- Lower reward for delays
- Penalty for high fuel consumption
- Reward for maximizing completed rides

4. Transition Probability (P)

Represents how the system changes after an action, considering factors such as traffic, new requests, and vehicle availability.

5. Discount Factor (γ)

Determines how much future rewards are considered compared with immediate rewards.

MDP Representation:

State $\rightarrow$ Action $\rightarrow$ Transition $\rightarrow$ Reward $\rightarrow$ New State

Conclusion:

MDP allows the ride-sharing or delivery system to make sequential decisions while optimizing long-term efficiency and customer satisfaction.

3. Compare model-free and model-based Reinforcement Learning approaches. Explain their working principles and how they handle uncertainty.

ANS: -

Feature	Model-Free RL	Model-Based RL
Environment Model	Not explicitly learned	Learns/uses environment model
Learning	Directly learns policy/value	Learns model and uses planning
Planning	Usually not required	Uses prediction and planning
Data Efficiency	Usually, lower	Usually, higher
Real Interaction	More interaction may be required	Can reduce real interaction
Uncertainty	Learns mainly from observed experience	Can explicitly model/predict uncertainty
Examples	DQN, PPO, SARSA	Dyna, MBPO

Model-Free RL

The agent directly learns the policy or value function from interactions with the environment.

Environment → Experience → Policy/Value Learning → Action

Model-Based RL

The agent learns an environment model that predicts future states and rewards. It uses this model for planning before selecting actions.

Environment → Learn Model → Predict Future → Plan → Action

Handling Uncertainty

- *Model-Free*: Handles uncertainty through repeated experience, exploration, and value estimation.
- *Model-Based*: Can represent uncertainty in predicted transitions and rewards and compare possible future outcomes.

Conclusion:

Model-free RL is simpler, while model-based RL can be more sample-efficient but depends on the accuracy of its environment model.

4. A recommendation system using Reinforcement Learning is underperforming. Explain how reward sparsity and hyperparameter tuning can be diagnosed and improved.

ANS:- If an RL-based recommendation system is underperforming, both reward sparsity and hyperparameter settings should be examined.

Reward Sparsity

Reward sparsity occurs when the agent receives meaningful feedback only occasionally.

For example, a recommendation system may receive a reward only when a user purchases an item.

How to Improve Reward Sparsity:

1. Reward Shaping: Provide intermediate rewards such as clicks, views, or time spent.
2. Reward Scaling: Normalize rewards to maintain stable learning.
3. Better Feedback: Include multiple user signals.
4. Exploration: Encourage the agent to try different recommendations.
5. Experience Replay: Reuse valuable experiences during training.

Hyperparameter Tuning

Important parameters include:

- Learning rate
- Discount factor (γ)
- Exploration rate (ϵ)
- Batch size
- Network architecture
- Replay-buffer size

Diagnostic Process:

Poor Performance → Analyze Rewards → Check Exploration → Tune Learning Rate → Test Hyperparameters → Evaluate Performance

Conclusion

Reward sparsity can be addressed using meaningful intermediate rewards, while systematic hyperparameter tuning improves convergence, stability, and recommendation quality.

5. If you are implementing an RL-based game-playing AI, describe how you would design the learning pipeline and ensure training stability.

ANS: - To develop an RL-based game-playing AI, the learning process should be organized into several stages.

Learning Pipeline

Game Environment → State Observation → Action Selection → Game Response
→ Reward → Experience Storage → Model Training → Policy Update

Steps:

1. Define Environment: Create game states, actions, rules, and rewards.
2. State Representation: Convert game information into a suitable input format.
3. Action Selection: Use an RL policy to select actions.
4. Reward Design: Give positive rewards for desirable actions and penalties for poor actions.
5. Experience Collection: Store state-action-reward transitions.
6. Model Training: Train the policy/value network using collected experiences.
7. Policy Update: Continuously improve the strategy.

Ensuring Training Stability:

- Use an appropriate learning rate.
- Apply experience replay where suitable.
- Use target networks for value-based methods such as DQN.
- Normalize inputs/rewards when necessary.
- Maintain a suitable exploration strategy.
- Monitor reward, loss, and convergence during training.
- Use evaluation games without exploration to measure actual performance.

Conclusion

A stable RL game-playing system requires a well-designed environment, meaningful rewards, controlled exploration, suitable hyperparameters, and continuous monitoring of training performance.

Code of Conduct:

I U. Somesh Kumar (192425019) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**U. Somesh Kumar****RUBRICS FOR CALCULATION**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately